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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Seventh Semester B.Tech (CBCS) Degree Examinations

December 2025

IS1105-1 – INTRODUCTION TO BIG DATA

(For Minor Degree)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. Which of the following best describes structured data?
A) Data that is stored in rows and columns in a database B) Data that is unorganized and without a predefined model
C) Data that comes only from sensors and IoT devices D) Data that cannot be analyzed using SQL
2. Which of the following is semi-structured data?
A) Video recordings B) JSON or XML files
C) Weather sensor values stored in a relational database D) Credit card numbers stored in a table
3. Which data type is easiest to analyze with traditional SQL-based systems?
A) Unstructured data B) Structured data
C) Semi-structured data D) Big data
4. The 3Vs model used to define Big Data includes:
A) Volume, Velocity, Variety B) Variety, Value, Visualization
C) Veracity, Validity, Visualization D) Volume, Visualization, Virtualization
5. Which of the following is NOT a characteristic of Big Data?
A) Volume B) Variety
C) Velocity D) Uniformity
6. One of the key goals of Big Data is to derive
A) More data storage capacity B) Automated transaction records
C) Meaningful insights and value from large datasets D) Only structured data analysis
7. The Sorting phase in Hadoop occurs:
A) At the Mapper output before shuffling B) At the Combiner level only
C) Before the data reaches Reducer (during Shuffle and Sort) D) After the final Reducer output is written to HDFS
8. In Scala, which symbol is used to define a function literal (anonymous function)?
A) -> B) =>
C) := D) ::
9. In Spark, DAG stands for:
A) Data Analysis Graph B) Data Aggregation Grid
C) Distributed Application Graph D) Directed Acyclic Graph
10. What is the key difference between transformations and actions in Spark?
A) Transformations trigger computation immediately, actions do not B) Transformations create new RDDs lazily, actions trigger execution of the DAG
C) Both are executed eagerly D) Actions can only be used with DataFrames, not RDDs



11. In Hadoop, HDFS stores data in units called:
 - A) Blocks
 - B) Files
 - C) Nodes
 - D) Racks
12. The default replication factor in HDFS is:
 - A) 2
 - B) 1
 - C) 3
 - D) 5
13. MapReduce consists of two main phases:
 - A) Partition and Combine
 - B) Mapper and Reducer
 - C) Shuffle and Sort only
 - D) Index and Query
14. The Partitioner in Hadoop is responsible for:
 - A) Assigning intermediate key-value pairs to reducers
 - B) Splitting data into blocks
 - C) Compressing input splits
 - D) Replicating data across DataNodes
15. Which tool in the Hadoop ecosystem is designed for scripting data flows?
 - A) Hive
 - B) Oozie
 - C) HBase
 - D) Pig
16. Which of the following is TRUE for external tables in Hive?
 - A) Data is deleted when the table is dropped
 - B) Only metadata is deleted when the table is dropped
 - C) External tables cannot use partitions
 - D) They do not support complex data types
17. Which of the following is true about DataFrames in Spark SQL?
 - A) They are immutable distributed collections of objects
 - B) They are schema-less
 - C) They represent structured data with schema
 - D) They cannot be queried with SQL
18. Which API combines the benefits of RDDs and DataFrames in Spark SQL?
 - A) DStream
 - B) GraphFrame
 - C) Thrift API
 - D) Dataset
19. What is the default web UI port for Spark Master in standalone mode?
 - A) 8080
 - B) 4040
 - C) 6066
 - D) 7077
20. In GraphX, a graph is represented using:
 - A) RDD of vertices and edges
 - B) Only adjacency matrix
 - C) Only adjacency list
 - D) SQL tables

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) List different digital data types and explain unstructured data, including its sources.	8	L*2	1	1
	b) Discuss various characteristics of big data.	8	L2	1	1,2
2.	a) Explain the Hadoop ecosystem in detail.	8	L2	1	1
	b) Explain how the objects are created and accessed in Scala with an example. Also specify whether the fields of an object are mutable or immutable.	8	L2	1	1,2
3.	a) Explain how you will match a value against a pattern using pattern matching in Scala.	8	L2	2	1,2
	b) Explain how Resilient Distributed Datasets are created.	8	L2	2	1,2
Unit – II					
4.	a) Illustrate how the Hadoop distributed file system (HDFS) can be designed.	8	L2	3	1,2
	b) Discuss various features of Hadoop.	8	L2	3	1,2

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| 5. | a) | Discuss about how Avro supports serialisation in Hadoop. | 8 | L2 | 3 | 1,2 |
| | b) | Illustrate how Pig Latin scripts are developed and tested. | 8 | L2 | 4 | 1,2 |
| 6. | a) | Explain various data types and file formats in Hive. | 8 | L2 | 4 | 1,2 |
| | b) | Explain different DML statements in Hive and write a HiveQL query that retrieves customer IDs and their total spending for orders placed in 2024, filtering for customers who spent over \$1000, and ordering the results by total spending in descending order. | 8 | L3 | 4 | 1,2 |

Unit – III

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| 7. | a) | Explain dataframes in Spark and write code to create a dataframe from structured data in Spark. | 8 | L3 | 5 | 1 |
| | b) | Explain the important features of the MongoDB database. | 8 | L2 | 5 | 1,2 |
| 8. | a) | Explain graph processing in Spark using GraphX. | 8 | L2 | 5 | 1 |
| | b) | Discuss different steps of using Apache Spark's MLlib and write code to illustrate the use of SVM. | 8 | L3 | 5 | 1,2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
